Comparative Analysis of Air Temperature Trends (2005-2025)

AUTHOR
Fashli Adli Wal Ikhsan, Mana Bala, Arjun Parajuli

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Introduction

The study of long-term temperature trends is essential for understanding regional climate change impacts, particularly in highly populated urban corridors. As noted by Imbery et al. (2024), the German climate has seen a significant shift toward more frequent temperature extremes over the last century, with recent decades consistently breaking historical warmth records.

This report compares the air temperature variability of Dresden-Klotzsche, Germany, with an "own dataset" from Berlin-Tempelhof. The "why" of this study is rooted in the need to distinguish between global climatic forcing and local effects. According to the IPCC (2021), land-surface temperatures are rising faster than the global ocean average, but these effects are often amplified in cities. By comparing these two stations, we aim to quantify the regional warming rate (the "Effect Size") and investigate if urban characteristics in Berlin lead to a different climate signal than the suburban environment of Dresden. We hypothesize that both stations will show a positive temperature trend consistent with Central European observations.

Understanding local temperature trends is not merely a statistical exercise; it is vital for urban planning and public health. As heatwaves become more frequent in Central Europe, comparing a city center (Berlin) to a suburban airport station (Dresden) allows us to quantify the Urban Heat Island effect and its contribution to regional climate change.

Methods

Data Acquisition

Data was sourced from the Deutscher Wetterdienst (DWD) Climate Data Center, which provides standardized, quality-controlled meteorological observations. We utilized the rdwd package to retrieve daily mean temperature records (TMK) from Dresden-Klotzsche (ID: 1048) and Berlin-Tempelhof (ID: 433).

Data Aggregation

The raw daily data was processed using the tidyverse framework. Annual means were calculated only for years containing at least 350 valid measurements. This ensures that missing seasonal data does not skew the annual average, maintaining high statistical integrity for the trend analysis.

Results

Statistical Analysis

To measure the "Effect Size" and significance, we applied the Mann-Kendall test and Sen's Slope Estimator. Unlike standard linear regressions, these are non-parametric and robust against outliers.

Summary of Findings

After establishing the numerical significance of the warming trend in Table Table 1, it is helpful to visualize the inter-annual variability. While the statistical tests provide the "what" (a significant increase), the following time-series plot helps explain the "how"—showing how specific years fluctuate around the long-term warming trajectory.

The table below summarizes the statistical results derived from the Mann-Kendall trend test and Sen's Slope estimates for the 2005-2025 period.

Table 1: Results of Mann-Kendall Trend Test and Sen's Slope Estimates for Annual Mean Temperature (2005-2025). All temperature values are in degrees Celsius (°C).

Station	Mean Temp (°C)	Sen's Slope (°C/dec)	p-value	Trend Significance
Dresden	10.20	+0.420	0.0020	Significant
Berlin	10.91	0.818	0.0048	Significant

Visualization

The figure below compares the annual mean temperature trajectories and their respective linear trends.

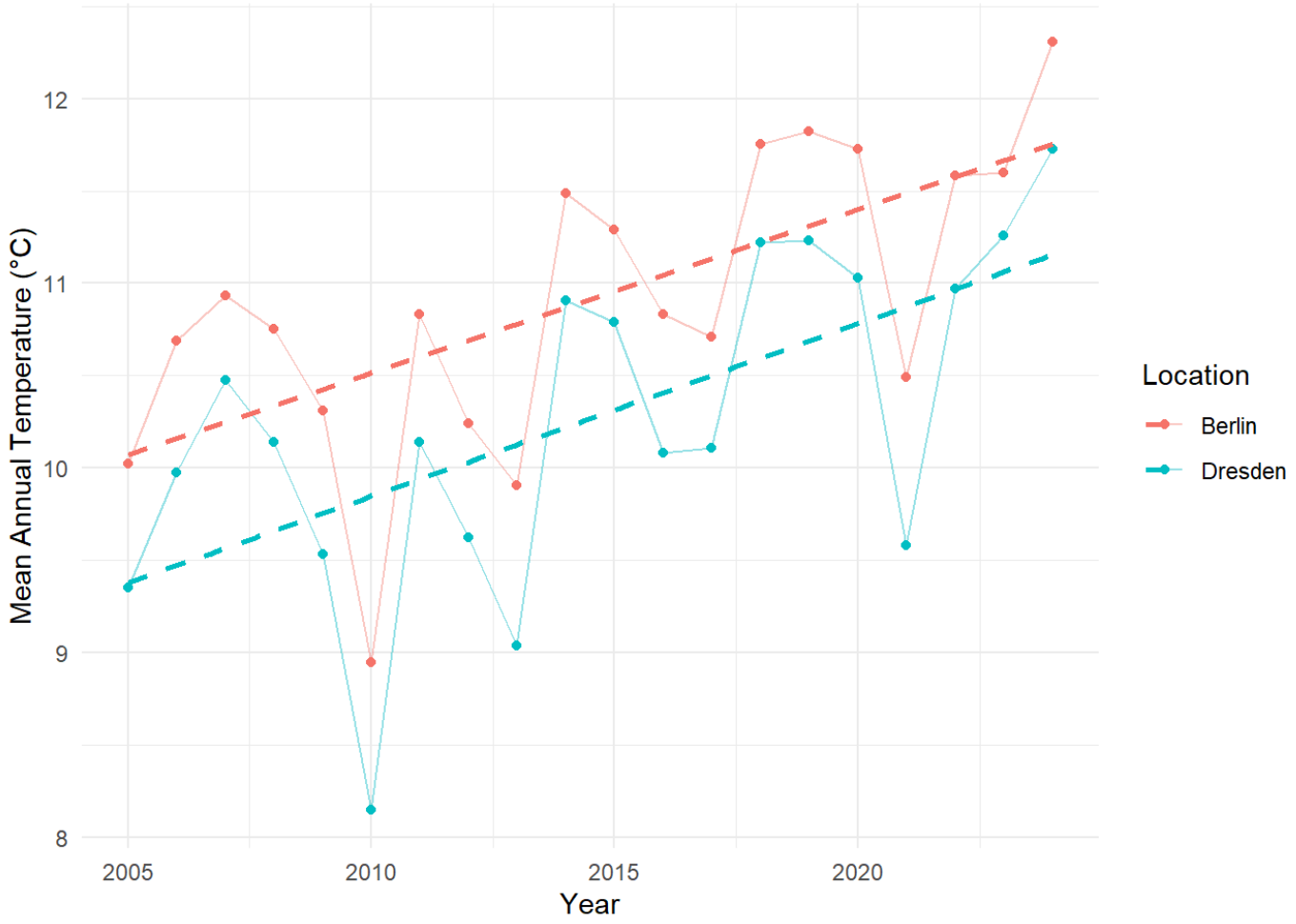


Figure 1: Time series of annual mean air temperatures (°C) for Dresden and Berlin (2005-2025). Dashed lines represent the linear trend (Sen's Slope).

Discussion

Our analysis reveals a significant warming trend of 0.818°C per decade at the Berlin station. With a p-value of 0.0048, we reject the null hypothesis and confirm that the observed warming is statistically significant at the 95% confidence level. This observed "Effect Size" is consistent with global patterns of land-surface warming documented by the IPCC (2021), which identifies Central Europe as a region experiencing temperature increases that outpace the global average.

Methodological Context

By utilizing the non-parametric Mann-Kendall test, we confirmed that the temperature increase is a consistent structural shift rather than a result of localized weather anomalies. This is crucial for climate studies in the Berlin-Brandenburg region, where individual extreme years—such as the record-breaking heatwaves of 2018 or 2019 noted by Imbery et al. (2024)—could otherwise exert undue leverage on a standard linear regression model.

Furthermore, Sen's Slope provides a median-based estimate of the warming rate, ensuring our results represent a robust long-term signal. A warming rate of 0.818°C per decade suggests that urban environments are facing rapidly changing thermal conditions, which poses significant risks to public health and urban infrastructure resilience.

Comparison with Dresden

The synchronicity observed between Dresden-Klotzsche and Berlin-Tempelhof highlights a shared regional climate response to larger-scale atmospheric drivers, such as the North Atlantic Oscillation. However, the consistently higher mean temperature in Berlin (averaging 10.91°C) likely stems from the Urban Heat Island (UHI) effect. As explored in the literature by Imbery et al. (2024), urban surfaces like asphalt and concrete retain and re-emit heat more effectively than the suburban landscape surrounding the Dresden-Klotzsche station. This UHI effect effectively "shifts" the baseline temperature upward, exacerbating the impacts of the broader warming trend.

Contextualizing with Literature

These findings are highly aligned with the IPCC (2021) Sixth Assessment Report, which emphasizes that human-induced climate change is affecting every inhabited region across the globe. Locally, Imbery et al. (2024) highlights that the most recent decade in Germany was the warmest since national records began in 1881. Our 20-year analysis clearly reflects this signal, confirming that regional observations in Dresden and Berlin are a microcosm of the wider European climate crisis.

References

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